

Commutative Algebra

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1. Rings and Ideals

1.1. Ideals, Quotient Rings

Proposition 1.1. Let $\mathfrak{a}$ be an ideal of A . The following is an one-to-one order-preserving correspondence between ideals $\mathfrak{b} \supseteq \mathfrak{a}$ of A and ideals $\bar{\mathfrak{b}}$ of $A/\mathfrak{a}$: $\mathfrak{b} \leftrightarrow q^{-1}(\bar{\mathfrak{b}})$. Here q is the surjective quotient map.

The homomorphism $f : A \rightarrow B$ induces an isomorphism $A/\ker(f) \simeq \text{im}(f)$.

1.2. Zero-divisors, Nilpotent Elements, Units

A *zero-divisor* in a ring A is an element x s.t. $\exists y \neq 0, xy = 0$. An *integral domain* is a ring with no zero-divisor.

$x \in A$ is *nilpotent* if $x^n = 0$ for some $n > 0$. A *unit* is an invertible element.

Proposition 1.2 Let A be a ring $\neq 0$. TFAE:

1. A is a field;
2. A only has trivial ideals;
3. If B is non-zero, $f : A \rightarrow B$ is a homomorphism, then f is injective.

1.3. Prime Ideals and Maximal Ideals

An ideal $\mathfrak{p}$ is *prime* if $\mathfrak{p} \neq (1)$ and $xy \in \mathfrak{p} \Rightarrow x \in \mathfrak{p} \vee y \in \mathfrak{p}$. An ideal $\mathfrak{m}$ is *maximal* if there is no $\mathfrak{m} \subsetneq \mathfrak{a} \subsetneq (1)$. Equivalently,

$\mathfrak{p}$ is prime $\Leftrightarrow A/\mathfrak{p}$ is an integral domain;

$\mathfrak{m}$ is maximal $\Leftrightarrow A/\mathfrak{m}$ is a field.

For a homomorphism $f : A \rightarrow B$, f^{-1} always maps a prime ideal to a prime ideal, but it doesn't necessarily map a maximal ideal to a maximal ideal. ($A = \mathbb{Z}, B = \mathbb{Q}, f^{-1}((0))$.)

Theorem. 1.3 Every $A \neq 0$ has at least one maximal ideal. (Under AC)

Proof.

Ideals $\neq (1)$ in A forms a nonempty poset under $\subset$. Every chain of ideals $\mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \dots$ is upper-bounded by $\mathfrak{a} = \bigcup \mathfrak{a}_k$. Take any maximal element by Zorn's lemma. ■

Corollary 1.4. If $\mathfrak{a} \neq (1)$ is an ideal of A , then $\exists$ a maximal ideal of A that contains $\mathfrak{a}$.

Corollary 1.5. Every non-unit of A is contained in a maximal ideal.

A ring with exactly one maximal ideal is called a *local ring*. A ring with finitely many maximal ideals is called *semi-local*.

Proposition 1.6.

1. If $\mathfrak{m} \neq (1)$ is an ideal of A s.t. every $x \in A \setminus \mathfrak{m}$ is a unit in A , then A is a local ring and $\mathfrak{m}$ is its maximal ideal.
2. If $\mathfrak{m}$ is a maximal ideal of A and $1 + \mathfrak{m} \subseteq A^\times$, then A is a local ring.

Proof.

1. Every ideal $\neq (1)$ consists of non-units and is contained in $\mathfrak{m}$.
2. Let $x \in A \setminus \mathfrak{m}$. Since $\mathfrak{m}$ is maximal, $(x) + \mathfrak{m} = (1)$, so $\exists t \in A, y \in \mathfrak{m}, xt + y = 1$. So $xt = 1 - y \in A^\times$ and so is x . This implies the premise of 1., hence the conclusion. ■

Note that the premise of 1. also implies 2.

Examples.

1. If f is irreducible in $k[x_1, x_2, \dots, x_n]$, then f is prime by unique factorization.
2. In $\mathbb{Z}$, every non-zero prime ideal is maximal since $\mathbb{Z}/(p)$ is a field. The fact holds for every *principal ideal domain*.

1.4. Nilradical and Jacobson Radical

Proposition 1.7. The set $N(A)$ of all nilpotent elements in A is an ideal, and $A/N(A)$ has no nilpotent element $\neq 0$. $N(A)$ is the *nilradical* of A .

Equivalently,

Proposition 1.8. $N(A)$ is the intersection of all the prime ideals of A .

Proof.

If $x^n = 0$, the definition of prime ideals shows that $x \in \mathfrak{p}$, for every prime ideal $\mathfrak{p}$.

On the other hand, suppose x isn't nilpotent. Consider the set of ideals $\mathfrak{a}$ s.t. $x^n \notin \mathfrak{a}, \forall n$. The set is non-empty since (0) is a member. By Zorn's lemma, this poset (under $\subset$) has a maximal element $\mathfrak{p}$. Let $y, z \notin \mathfrak{p}$. Since $(y) + \mathfrak{p}$ and $(z) + \mathfrak{p}$ strictly contain $\mathfrak{p}$, they do not belong to the poset, hence

$$x^m \in (y) + \mathfrak{p}, x^n \in (z) + \mathfrak{p}.$$

Therefore $x^{m+n} \in (yz) + \mathfrak{p}$ and $(yz) + \mathfrak{p}$ doesn't belong to the poset. Thus $yz \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime. ■

The *Jacobson radical* $J(A)$ is the intersection of the maximal ideals of A .

Proposition 1.9. $x \in J(A) \Leftrightarrow 1 - xy \in A^\times, \forall y \in A$.

Proof.

$\Rightarrow$: If $(1 - xy) \neq (1)$, then by (1.5) it is contained in some maximal $\mathfrak{m}$, which x belongs to. This implies $1 \in \mathfrak{m}$, contradiction.

$\Leftarrow$: Suppose $x \notin \mathfrak{m}$ for some maximal $\mathfrak{m}$. We have $(x) + \mathfrak{m} = (1)$ hence $u + xy = 1$ for some $u \in \mathfrak{m}$ and $y \in A$. Hence $1 - xy \in \mathfrak{m}$ and therefore is not a unit. ■

1.5. Operations on Ideals

Those include $\mathfrak{a} + \mathfrak{b} = \{x + y \mid x \in \mathfrak{a}, y \in \mathfrak{b}\}$, $\mathfrak{a}\mathfrak{b} = (xy)(x \in \mathfrak{a}, y \in \mathfrak{b})$, $\mathfrak{a} \cap \mathfrak{b}$. The intersection can be defined on any family of ideals.

The 3 operations are commutative and associative. We also have the *distributive law*

$$\mathfrak{a}(\mathfrak{b} + \mathfrak{c}) = \mathfrak{a}\mathfrak{b} + \mathfrak{a}\mathfrak{c}.$$

$\cap$ and $+$ are distributive in $\mathbb{Z}$ but it is not the case in general. The *modular law* states

$$\mathfrak{b} \subseteq \mathfrak{a} \vee \mathfrak{c} \subseteq \mathfrak{a} \Rightarrow \mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) = \mathfrak{a} \cap \mathfrak{b} + \mathfrak{a} \cap \mathfrak{c}.$$

In general, we only have $\supseteq$. (Take $A = F[x, y]$, let $\mathfrak{a} = (x)$, $\mathfrak{b} = (y)$, $\mathfrak{c} = (x + y)$.)

In $\mathbb{Z}$, we also have $(\mathfrak{a} + \mathfrak{b})(\mathfrak{a} \cap \mathfrak{b}) = \mathfrak{a}\mathfrak{b}$, but in general only $LHS \subseteq RHS$. $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b}$, so when $\mathfrak{a} + \mathfrak{b} = (1)$, $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b}$.

$\mathfrak{a}, \mathfrak{b}$ are said to be *coprime* if $\mathfrak{a} + \mathfrak{b} = (1)$.

The *direct product* of rings are defined with componentwise operations. The projections are surjective ring homomorphisms.

Let $\mathfrak{a}_{[n]}$ be ideals of A . Define a homomorphism

$$\phi : A \rightarrow \prod_{i=1}^n (A/\mathfrak{a}_i)$$

by $\phi(x) = (x + \mathfrak{a}_1, x + \mathfrak{a}_2, \dots, x + \mathfrak{a}_n)$.

Proposition 1.10.

1. If $\mathfrak{a}_i \perp \mathfrak{a}_j (\forall i < j)$, then $\prod \mathfrak{a}_i = \bigcap \mathfrak{a}_i$.
2. ϕ surjective $\Leftrightarrow \forall i < j, \mathfrak{a}_i \perp \mathfrak{a}_j$.
3. ϕ injective $\Leftrightarrow \bigcap \mathfrak{a}_i = (0)$.

Proof.

1. Induction on n . $n = 2$ is dealt with. For $n > 2$, suppose the result is true for $n - 1$, let $\mathfrak{b} = \prod_{i \leq n-1} \mathfrak{a}_i = \bigcap_{i \leq n-1} \mathfrak{a}_i$. Suppose $x_i + y_i = 1 (x_i \in \mathfrak{a}_i, y_i \in \mathfrak{a}_n, 1 \leq i \leq n - 1)$. Then

$$\prod_{i \leq n-1} x_i = \prod_{i \leq n-1} (1 - y_i) \equiv 1 \pmod{\mathfrak{a}_n}.$$

Hence $\mathfrak{a}_n + \mathfrak{b} = 1$, so the conclusion holds.

2. $\Rightarrow$: To show $\mathfrak{a}_1 + \mathfrak{a}_2 = 1$, take $x \in A$ s.t. $\phi(x) = (1, 0, \dots, 0)$, then $1 = (1 - x) + x \in \mathfrak{a}_1 + \mathfrak{a}_2$.

$\Leftarrow$: Use something similar to the proof of CRT.

3. Trivial by definition. ■

Proposition 1.11.

1. Let $\mathfrak{p}_{[n]}$ be prime ideals, $\mathfrak{a} \subseteq \bigcup \mathfrak{p}_i$. Then $\exists i, \mathfrak{a} \subseteq \mathfrak{p}_i$.
 2. Let $\mathfrak{p} \supseteq \bigcap \mathfrak{a}_i$. Then $\exists i, \mathfrak{p} \supseteq \mathfrak{a}_i$. Moreover, if $\mathfrak{p} = \bigcap \mathfrak{a}_i$, then $\exists i, \mathfrak{p} = \mathfrak{a}_i$.

Proof.

1. Induction on n . Assume for every $\mathfrak{a}$ and $\mathfrak{p}$,

$$\mathfrak{a} \not\subseteq \mathfrak{p}_i (1 \leq i \leq n - 1) \Rightarrow \mathfrak{a} \not\subseteq \bigcup_{i=1}^{n-1} \mathfrak{p}_i.$$

This is true for trivial case. If $n > 1$, then $\forall i, \exists x_i \in \mathfrak{a}$ s.t. $\forall j \neq i, x_i \notin \mathfrak{p}_j$. If $\exists i, x_i \notin \mathfrak{p}_i$ then we are done. Otherwise,

$$y = \sum_{i=1}^n \prod_{j \neq i} x_j \in \mathfrak{a} \setminus \bigcup_{i=1}^n \mathfrak{p}_i.$$

2. Suppose $x_i \in \mathfrak{a}_i \setminus \mathfrak{p}$. Then $\prod x_i \in \prod \mathfrak{a}_i \subseteq \bigcap \mathfrak{a}_i$. Since $\mathfrak{p}$ is prime, $\prod x_i \notin \mathfrak{p}$. Hence $\mathfrak{p} \not\supseteq \bigcap \mathfrak{a}_i$. The corollary is trivial. ■

The *ideal quotient* of $\mathfrak{a}, \mathfrak{b}$ is

$$(\mathfrak{a} : \mathfrak{b}) = \{x \in A : x\mathfrak{b} \subseteq \mathfrak{a}\}$$

which is an ideal.

$(0 : \mathfrak{b})$ is called the *annihilator* of $\mathfrak{b}$ or $\text{Ann}(\mathfrak{b})$. The zero-divisors in A is the union of all annihilators of non-zero elements.

Exercise 1.12.

- $(\mathfrak{a} : \mathfrak{b})\mathfrak{b} \subseteq \mathfrak{a} \subseteq (\mathfrak{a} : \mathfrak{b})$.
- $((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c}) = (\mathfrak{a} : \mathfrak{b}\mathfrak{c}) = ((\mathfrak{a} : \mathfrak{c}) : \mathfrak{b})$.
- $(\bigcap_i \mathfrak{a}_i : \mathfrak{b}) = \bigcap_i (\mathfrak{a}_i : \mathfrak{b})$.
- $(\mathfrak{a} : \sum_i \mathfrak{b}_i) = \bigcap_i (\mathfrak{a} : \mathfrak{b}_i)$.

The *radical* of an ideal $\mathfrak{a}$ is

$$\sqrt{\mathfrak{a}} = \{x \in A : x^n \in \mathfrak{a} \text{ for some } n > 0\}.$$

We have $\sqrt{\mathfrak{a}} = q^{-1}(N(A/\mathfrak{a}))$ where $q : A \rightarrow A/\mathfrak{a}$ is the quotient map, hence the radical is always an ideal.

Exercise 1.13.

1. $\sqrt{\mathfrak{a}} \supseteq \mathfrak{a}$.
2. $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$.
3. $\sqrt{\mathfrak{a}\mathfrak{b}} = \sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}} \cap \sqrt{\mathfrak{b}}$.
4. $\sqrt{\mathfrak{a}} = (1) \Leftrightarrow \mathfrak{a} = 1$.
5. $\sqrt{\mathfrak{a} + \mathfrak{b}} = \sqrt{\sqrt{\mathfrak{a}} + \sqrt{\mathfrak{b}}}$.
6. $\mathfrak{p}$ is prime implies $\sqrt{\mathfrak{p}^n} = \mathfrak{p}, \forall n > 0$.

Proposition 1.14. $\sqrt{\mathfrak{a}}$ is the intersection of $\mathfrak{p} \supseteq \mathfrak{a}$.

We can define radical on any subset E , but $\sqrt{E}$ is not necessarily an ideal.

Proposition 1.15. Let D be the zero-divisors of A . Then $D = \sqrt{D} = \sqrt{\bigcup_{x \neq 0} \text{Ann}(x)} = \bigcup_{x \neq 0} \sqrt{\text{Ann}(x)}$.

Proposition 1.16. $\sqrt{\mathfrak{a}} \perp \sqrt{\mathfrak{b}} \Leftrightarrow \mathfrak{a} \perp \mathfrak{b}$.

1.6. Extension and Contraction

Fix a homomorphism $f : A \rightarrow B$. If $\mathfrak{a}$ is an ideal of A , then $f(\mathfrak{a})$ is not always an ideal. (Consider $A = \mathbb{Z}, B = \mathbb{Q}$.) Define the *extension* $\mathfrak{a}^e = (f(\mathfrak{a}))$ which is an ideal of B . If $\mathfrak{b}$ is an ideal of B , then $f^{-1}(\mathfrak{b})$ is always an ideal of A , called the *contraction* $\mathfrak{b}^c$ of $\mathfrak{b}$.

Contraction of a prime ideal is always prime. This doesn't hold for extensions.

f can be factored as follows:

$$A \xrightarrow{i} f(A) \xrightarrow{j} B.$$

From (1.1), i is simple: ideals of $f(A) \xrightarrow{1:1} \text{ideals of } A \supseteq \ker(f)$. However, j is in general complicated.

Example.

Consider $\mathbb{Z} \rightarrow \mathbb{Z}[i]$. $\mathbb{Z}[i]$ is a principal domain. Consider prime ideal (p) of $\mathbb{Z}$.

1. $(2)^e = ((1+i)^2)$ which is the square of a prime ideal.
2. $p \equiv 1 \pmod{4} \Rightarrow (p)^e$ is the product of two distinct prime ideals.
3. $p \equiv 3 \pmod{4} \Rightarrow (p)^e$ is prime in $\mathbb{Z}[i]$.

Here 2. is a non-trivial result and is equivalent to *Fermat's square theorem*.

Proposition 1.17.

1. $\mathfrak{a} \subseteq \mathfrak{a}^{ec}, \mathfrak{b} \supseteq \mathfrak{b}^{ce}$.
2. $\mathfrak{a}^e = \mathfrak{a}^{ece}, \mathfrak{b}^c = \mathfrak{b}^{cec}$.
3. Let C be the image of $*^c$ and E be the image of $*^e$. Then $C = \{\mathfrak{a} \mid \mathfrak{a}^{ec} = \mathfrak{a}\}, E = \{\mathfrak{b} \mid \mathfrak{b}^{ce} = \mathfrak{b}\}$, and $\mathfrak{a} \mapsto \mathfrak{a}^e, \mathfrak{b}^c \mapsto \mathfrak{b}$ gives a bijection $C \xrightarrow{1:1} E$.

Exercise 1.18.

1. $(\mathfrak{a}_1 + \mathfrak{a}_2)^e = \mathfrak{a}_1^e + \mathfrak{a}_2^e, (\mathfrak{b}_1 + \mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c$.
2. $(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e, (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c = \mathfrak{b}_1^c \cap \mathfrak{b}_2^c$.
3. $(\mathfrak{a}_1 \mathfrak{a}_2)^e = \mathfrak{a}_1^e + \mathfrak{a}_2^e, (\mathfrak{b}_1 \mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c$.
4. $(\mathfrak{a}_1 : \mathfrak{a}_2)^e \subseteq (\mathfrak{a}_1^e : \mathfrak{a}_2^e), (\mathfrak{b}_1 : \mathfrak{b}_2)^c \subseteq (\mathfrak{b}_1^c : \mathfrak{b}_2^c)$.
5. $\sqrt{\mathfrak{a}}^e \subseteq \sqrt{\mathfrak{a}^e}, \sqrt{\mathfrak{b}}^c = \sqrt{\mathfrak{b}^c}$.

1.7. The Prime Spectrum of a Ring

Let X be the set of prime ideals in A . $\forall E \subseteq A$, let $V(E) = \{\mathfrak{p} \in X : E \subseteq \mathfrak{p}\}$. We have the following properties:

1. Let $\mathfrak{a} = (E)$. Then $V(E) = V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$.
2. $V(0) = X$, $V(1) = \emptyset$.
3. $V(\bigcup_{i \in I} E_i) = \bigcap_{i \in I} V(E_i)$.
4. $\forall \mathfrak{a}, \mathfrak{b}$ of A , $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$.

Thus the family of sets $\{V(E)\}$ forms the closed sets of a topology, which we refer to as the *Zariski topology*. X is called the *prime spectrum* of A or $\text{Spec}(A)$.

Examples.

1. $\text{Spec}(\mathbb{Z}) = \{0, (p)\}$ and $V(\cdot)$ consists of the finite non-zero subsets of $\text{Spec}(\mathbb{Z})$ and the entire $\text{Spec}(\mathbb{Z})$.
2. $\text{Spec}(F) = \{0, F\}$.
3. $\text{Spec}(\mathbb{C}[x]) = \{0, (x - \alpha)_{\alpha \in \mathbb{C}}\}$, $V(\cdot)$ is similar to 1.
4. $\text{Spec}(\mathbb{R}[x]) = \left\{0, (x - \alpha)_{\alpha \in \mathbb{R}}, (x^2 + bx + c)_{\Delta < 0}\right\}$.
5. $\text{Spec}(\mathbb{Z}[x]) = \{0, (p), (p, g(x)), (f(x))\}$ where f is a primitive irreducible polynomial, g is irreducible in $\mathbb{F}_p[x]$ (this is beyond the scope of this chapter).

Define $X_f = (V(f))^c \in \text{Spec}(A)$ where $f \in A$. Then X_f is an open set. We have

1. $X_f \cap X_g = X_{fg}$.
2. $X_f = \emptyset \Leftrightarrow f$ is nilpotent.
3. $X_f = X \Leftrightarrow f \in A^\times$.
4. $X_f = X_g \Leftrightarrow \sqrt{(f)} = \sqrt{(g)}$.
- 5.

Bibliography